Lecture Slip, CS 235
Nov 5, 2026, Thursday
Shunting-Yard Algorithm

Name:	
EmpID:	

1. **Three notations.** Consider the infix expression `6 + 2 * (5 - 1)`.

   (a) Write it in **postfix** and in **prefix**.

   (b) Evaluate the postfix form with one stack. Write the stack (bottom to top) just before the `+` is processed, and the final value.

2. **Operand order.** Evaluate the postfix expression `20 6 2 - /`.

   (a) Show the stack (bottom to top) after each token.

   (b) In `evalPostfix`, `b` is popped first and `a` second. Suppose a student reverses the operands of **both** `-` and `/` (computing `b - a` and `b / a`). What result does the code produce?

3. **One character matters.** The operator loop in `toPostfix` uses `prec(ops.top()) >= prec(t)`. A student changes it to `prec(ops.top()) > prec(t)`.

   (a) What postfix does the changed code produce for `16 - 4 - 2`, and what does it evaluate to? What is the correct result?

   (b) Which property of `-` does the `>=` test protect?

4. **Shunting-yard trace.** Convert `8 - 3 * 2 + 1` to postfix. Fill in the table. In "Action", record the action or sequence of actions (output, push, pop); an incoming operator may cause several stacked operators to be popped. The last row is the end of the input.

Step	Token	Action	Output	Stack (bottom to top)
1	`8`			
2	`-`			
3	`3`			
4	`*`			
5	`2`			
6	`+`			
7	`1`			
8	*end*			

Which row shows the rule "pop operators of **higher or equal** precedence"? What value does the postfix result give?

5. **Pace survey.** Please help me set the pace of the course. Circle one.

Too **slow** (I can go faster)	**About right** (normal pace)	Too **fast** (please slow down)

6. **Topics to clarify.** Check **all** topics you would like me to explain again next class.

- ☐ Infix, prefix, and postfix notation
- ☐ Evaluating postfix with a stack
- ☐ Which operand is popped first
- ☐ Tokenizing (multi-digit numbers)
- ☐ Shunting-yard: precedence and ties (`>=`)
- ☐ Shunting-yard: parentheses
- ☐ The code: `tokenize`, `evalPostfix`, `toPostfix`
- ☐ Other: _____

7. Any other questions about today's class? If so, please provide details. Thank you.